

Felipe Gutierrez

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[Link to Portfolio](#)

CAREER SUMMARY

- Third-year Mechanical Engineering (Honours) student at UTS with hands-on experience in mechanical design, structural analysis, and SolidWorks CAD.
- Focused on mechanical design and physical systems engineering, with a drive to apply this knowledge to real engineering problems.

TECHNICAL SKILLS

- **SolidWorks (Proficient):** Designed and simulated mechanical systems across multiple subjects, including CFD analysis with SolidWorks Flow Simulation.
- **MATLAB, Python, C++ (Basic):** Applied control and data analysis scripts in Introduction to Mechatronics.
- **Mechatronics Integration (Basic):** Combined mechanical, electrical, and software components in a lab setting to enhance existing systems.
- **Microsoft Office (Proficient):** Skilled in Word, Excel, and PowerPoint for data analysis, documentation, and presentations.
- **Languages:** English (Bilingual proficiency) and Spanish (Bilingual proficiency).

EDUCATION

Bachelor of Mechanical Engineering (Honours)

University of Technology Sydney

Aug 2024 – Nov 2028

EXPERIENCE

Warrane College – Residential Tutor

Sydney, Australia

Feb 2026 – Present

- Provided pastoral care and mentorship to residents, supporting their transition into university life.
- Promoted student wellbeing through regular check-ins, informal counselling, and early intervention.
- Fostered an inclusive residential environment across diverse cultural and academic backgrounds.
- Planned and delivered weekly floor events to encourage engagement and community culture.
- Offered peer guidance and conflict support, reinforcing respectful and responsible college living.

Cake Shop and Coffee, Pasteleria Alvarez Carmona

Coquimbo, Chile

Nov 2018 – Apr 2022

- Worked as a waiter in a customer service-oriented cafe, delivering barista and table service.
- Assisted with inventory management, restocking and organising products to meet demand.

PROJECTS

2026 Weir Warman Design & Build Competition

41059 Mechanical Design Fundamentals Studio 1 | University of Technology Sydney

Autumn 2026 – Present

- Designed and built an autonomous robot to collect and deposit 3D-printed rocks within a 120-second limit as part of a five-person team in the 39th Weir Warman Competition.
- Served as Research Lead and Quality & Safety Lead, producing risk assessments and Safe Work Method Statements (SWMS).
- Produced AS1100-compliant engineering drawings and 3D CAD models using SolidWorks 2025.
- Applied morphological tables, weighted decision matrices, and iterative design reviews to select the final robot concept.

Counter-Flow Heat Exchanger Design and Analysis

41057 Thermofluids A | University of Technology Sydney

Autumn 2025

- Designed and modelled multiple double-pipe heat exchanger configurations in SolidWorks, including an octagonal counter-flow variant, and produced a 3D-printed prototype.
- Ran CFD simulations using SolidWorks Flow Simulation to analyse temperature distribution and heat transfer performance.
- Applied the NTU method to analytically validate simulation results and compare thermal effectiveness across design variants.
- Synthesised experimental, computational, and theoretical findings into a detailed technical report.

Vertical Axis Wind Turbine – Design, Fabrication and Testing

Spring 2025

41053 Materials and Manufacturing Engineering A | University of Technology Sydney

- Designed, fabricated, and tested a six-blade VAWT in a four-person team, generating measurable electrical output at low wind speeds.
- Designed 3D-printed PETG central connectors in SolidWorks with dimensional tolerances to ensure accurate fit with aluminium and steel components.
- Fabricated components using UTS workshop equipment (guillotine, slip roller, drill press) and assembled using bolted joints.
- Tested on a roller-driven rig, recording rotational speed and electrical output under simulated wind conditions.

VOLUNTEERING

Warrane College Volunteering – Little Sisters Nursing Home

Feb 2024 – Present

Randwick, NSW

- Assisted with weekly gardening and maintenance tasks to maintain a welcoming environment.
- Collaborated with volunteers to organise supplies and set up communal areas as needed.
- Built interpersonal and empathy skills by regularly interacting with and offering companionship to elderly residents.
- Committed to a consistent Saturday schedule, demonstrating reliability and community spirit.

EXTRA-CURRICULAR EXPERIENCE

President, Chess Club at Warrane College

Jan 2024 – Dec 2024

Sydney, Australia

- Organised weekly sessions and inter-college tournaments, fostering a supportive environment for all skill levels.
- Demonstrated leadership and event management skills while coordinating club activities and promoting engagement.

UNSW Intercollege Cup Competitor, Warrane College

Apr 2023 – Present

Sydney, Australia

- Represented Warrane College in OzTag, football, and basketball, demonstrating teamwork, resilience, and sportsmanship.
- Balanced athletic commitments with academic responsibilities, demonstrating time management and dedication.

Class President, Year 12

Feb 2022 – Dec 2022

Colegio Ingles Catolico La Serena, Chile

- Managed class funds and organised key activities (graduation events, fundraisers), coordinating with staff and peers.
- Strengthened leadership and communication skills by mediating between students, faculty, and administration.

REFEREES

Available upon request.